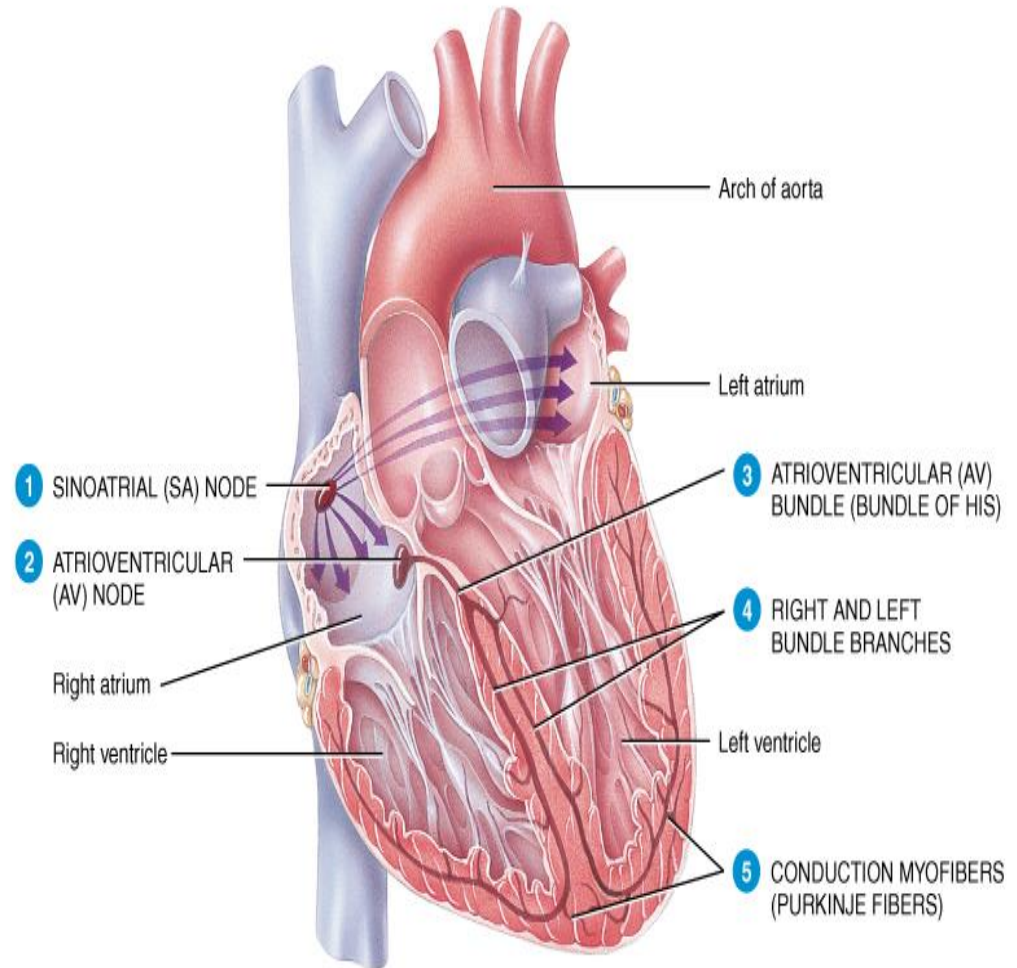


Electrophysiology of the Heart

Properties of Cardiac Muscles

- Cardiac muscles have 4 physiological properties, which are essential for the function of the heart as the central pumping organ.
- These are
 - ✓ Autorythmicity
 - ✓ Excitability
 - ✓ Conductivity
 - ✓ Contractility



Automaticity

- Cardiac function is *not fully dependent on intact nervous pathways*.
- It continues to beat after full denervation because of the intrinsic properties it has.
- **Automaticity** is the ability to start pace making is most highly developed in nodal tissues but can be observed in any piece of cardiac tissue.
 - Inherent rates:
 - SA node =70-80 bpm, AV node =60-65 bpm
 - Atria = 60-65 bpm, Ventricles =30-40 bpm

Autorythmicity

- The source of this electrical activity is from the network of specialized cardiac muscle fibres **called pacemaker cells or autorhythmic fibres** found on the right atrium.
 - They control and coordinate the rate of the heart beat.
- Each heartbeat begins with an action potential generated at a pacemaker cells in the right atria.
- The conducting system then propagates and distributes this **electrical impulse to stimulate contractile cells of the heart.**

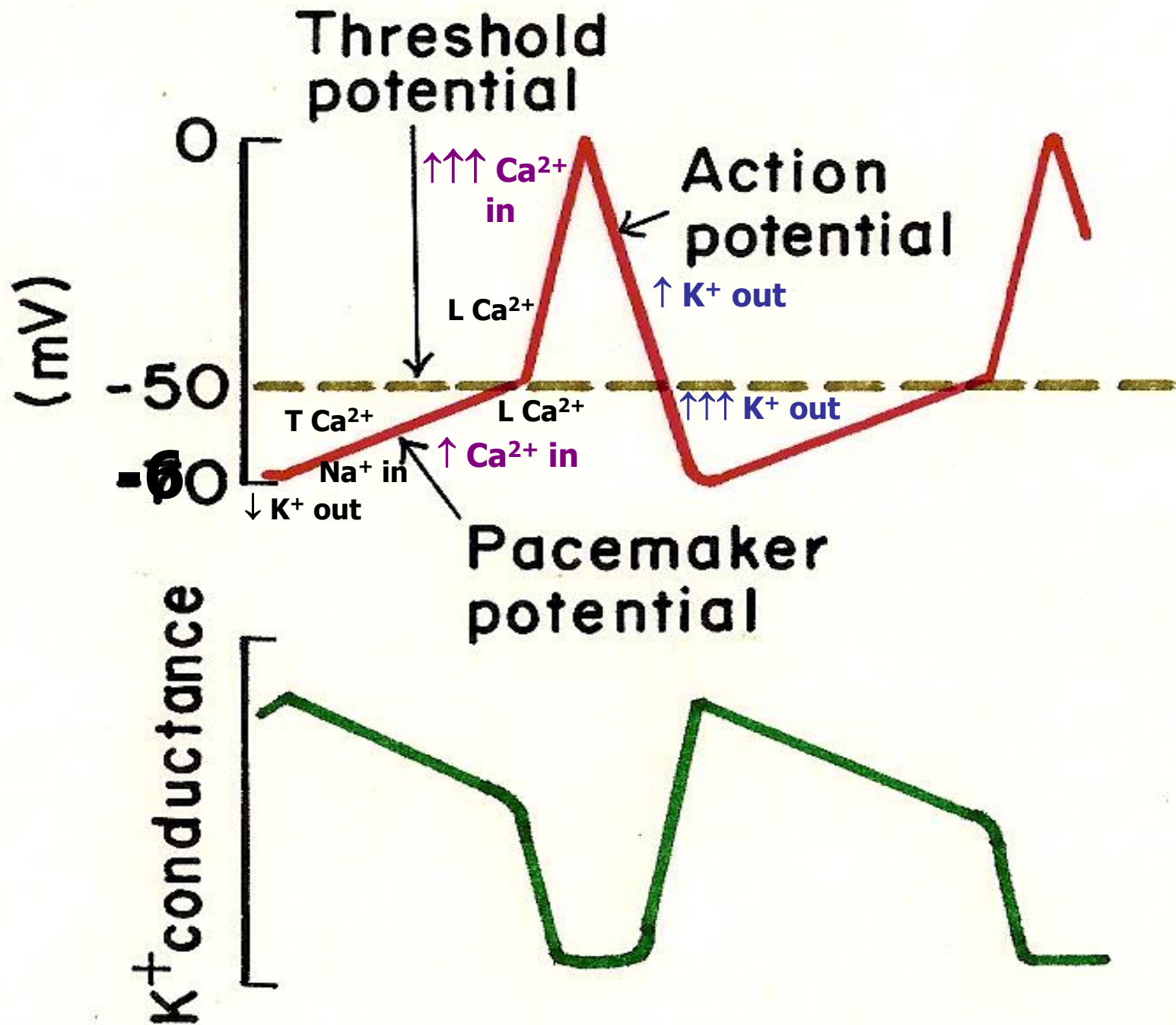
Pacemaker/Sinoatrial (SA) Potentials

- The RMP of the **sinus nodal** fibre is between -55 to -60 mv.
- Why these cells contain less negativity in the ICF?
 - The cell membranes of the sinus fibres are *naturally leaky to sodium and calcium ions.*
 - Opening of the fast sodium channels allow the rapid *influx of positive sodium ions to the interior of the fiber.*
 - Influx of positively charged sodium ions causes *a slow rise in the resting membrane potential in the positive direction.*

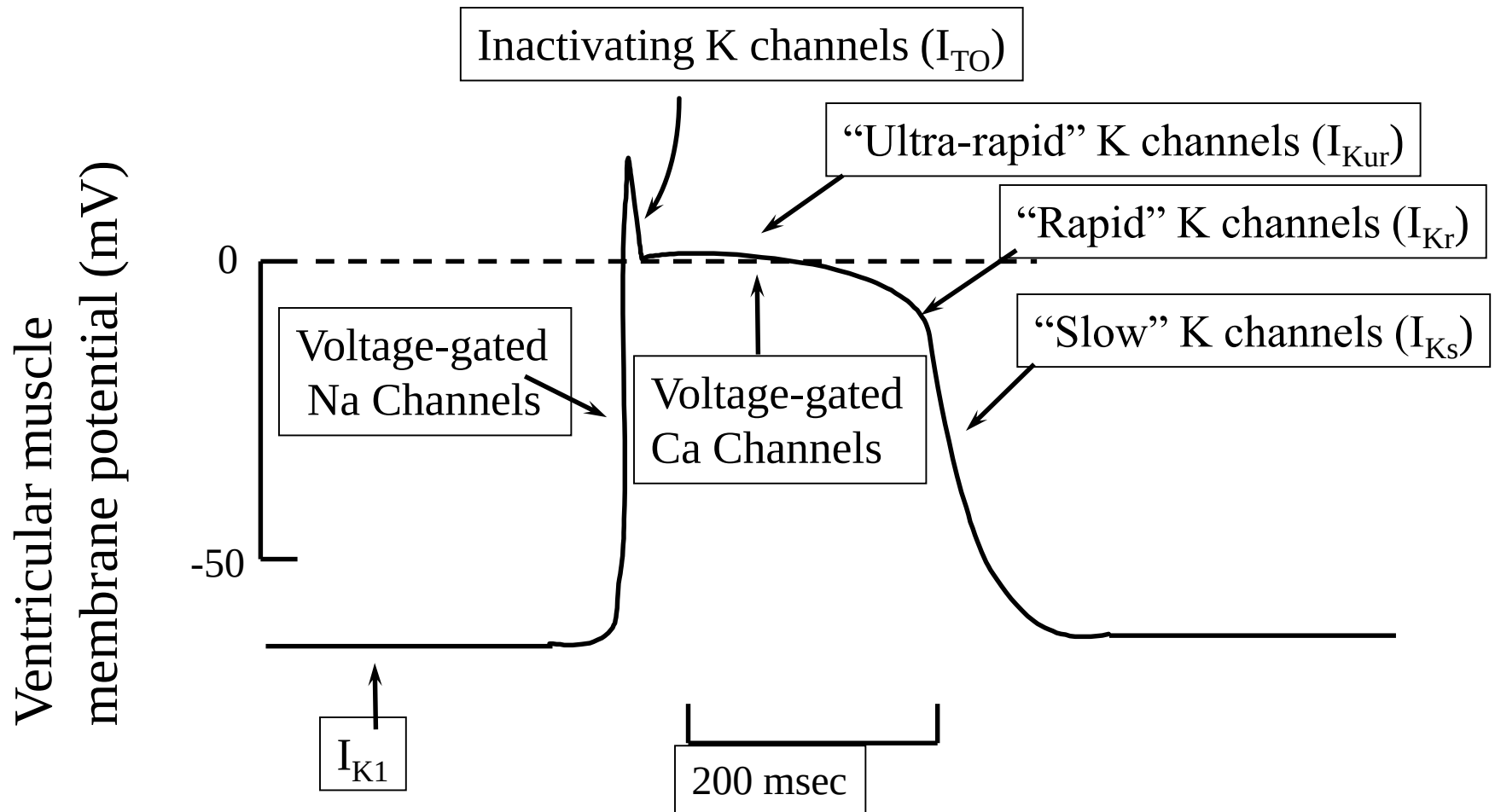
Pacemaker/Sinoatrial (SA) Potentials

- Slowly opened activated **calcium channels** will cause the opening of very much calcium ion channels and leads to the **generation of action potential**.
- Therefore, basically, *the inherent leakiness of the sinus nodal fibres to sodium and calcium ions causes their self-excitation*.
- SA-nodal discharge decreases with *age*.
- It can be calculated using a formula.
 - Intrinsic SA-nodal rhythm = $118.1 - (0.57 \times \text{Age})$.

Membrane Potential (mV)



Ion Channels in Ventricular Muscle



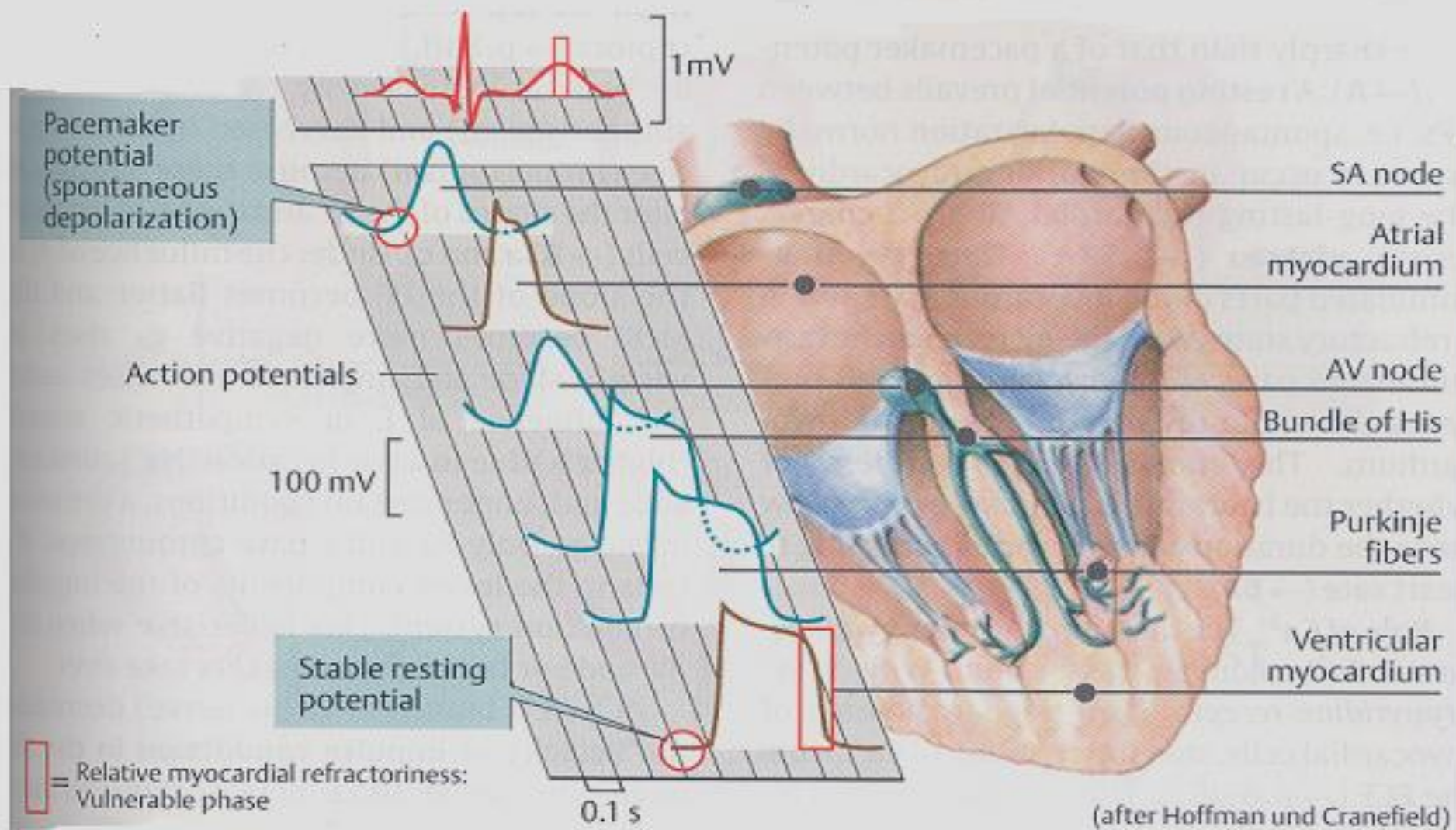
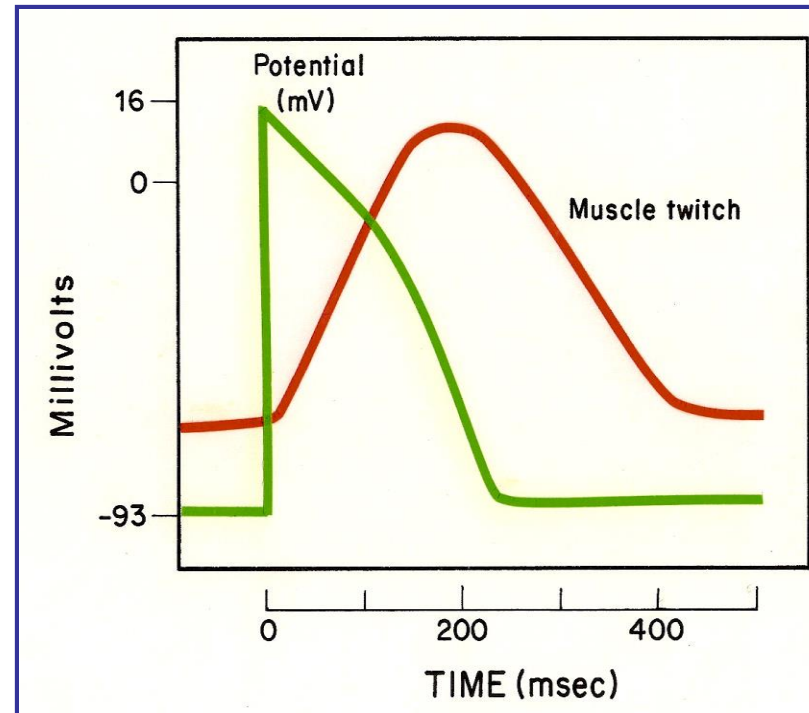


Fig..Cardiac excitation

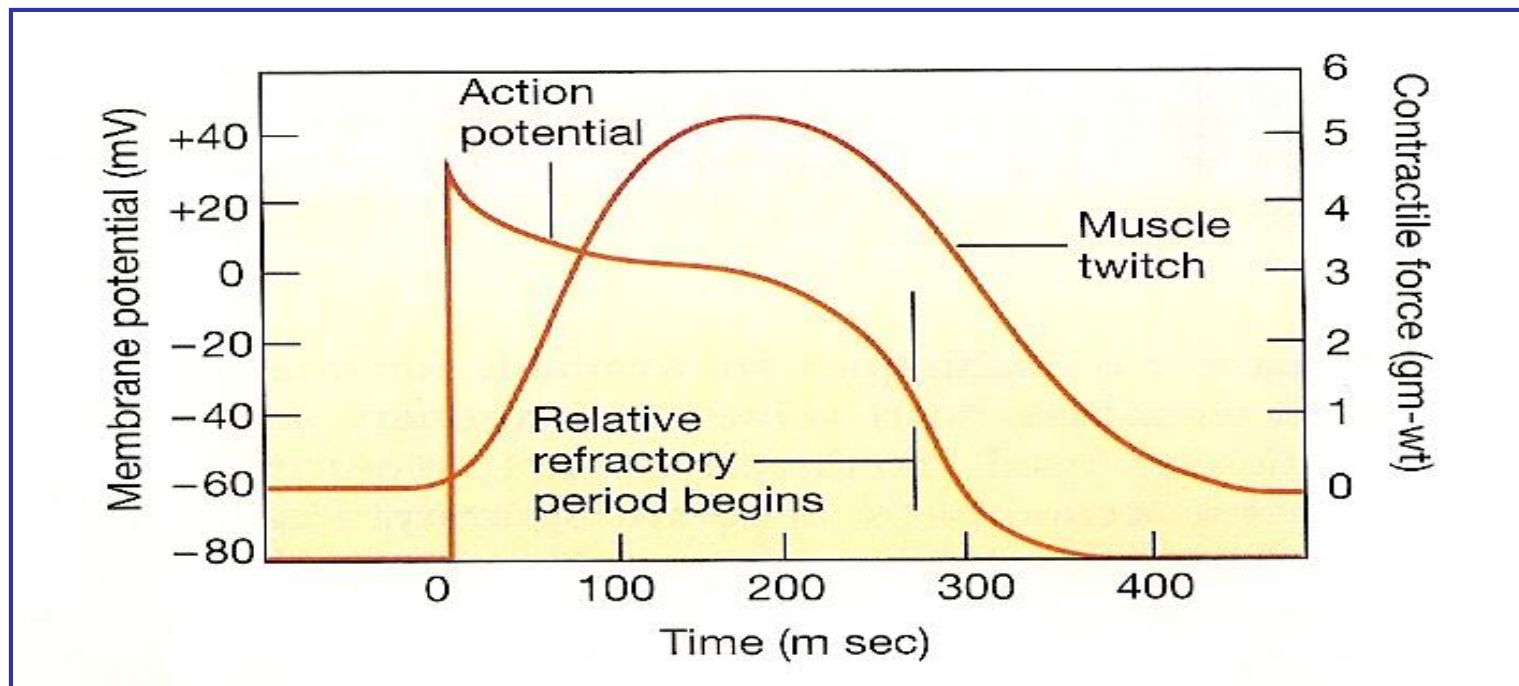
Relation between the Action Potential and the Mechanical Response

- The mechanical response consists of contraction (systole) & relaxation (diastole).
- Cardiac muscle begins to contract few milliseconds after the AP begins, & continues to contract until few milliseconds after the AP ends.
- Duration of contraction:
 - ≈ 0.2 sec in arial muscle, &
 - ≈ 0.3 sec in ventricular muscle.



Relation between the action potential and the mechanical response

- Diastole begins at the end of the plateau.
- Rapid repolarization is completed at about the middle of diastole.



Excitation-Contraction Coupling in Heart Muscle

- ❖ is the mechanism by which AP causes myofibrils of cardiac muscle to contract.
- ♥ When AP passes over cardiac muscle membrane, it also spread to interior of cardiac muscle fiber along membranes of transverse (T) tubules.
- ♥ Depolarization of myocardial cell stimulates opening of Ca^{2+} channels in sarcolemma.
 - Ca^{2+} diffuses down gradient into cell through T tubules.
 - Stimulates opening of Ca^{2+} -release channels in SR.
 - Ca^{2+} binds to troponin & stimulates contraction (same mechanisms as in skeletal muscle).

Excitation-Contraction Coupling

♥ At the end of plateau of cardiac AP, i.e. during repolarization,

■ Ca^{2+} in sarcoplasm is rapidly & actively transported & pumped out of the cell via a *$\text{Na}^+ - \text{Ca}^{2+}$ exchanger*, back into both SR & T tubules.

■ Resulting in cessation of the contraction until new AP occurs.

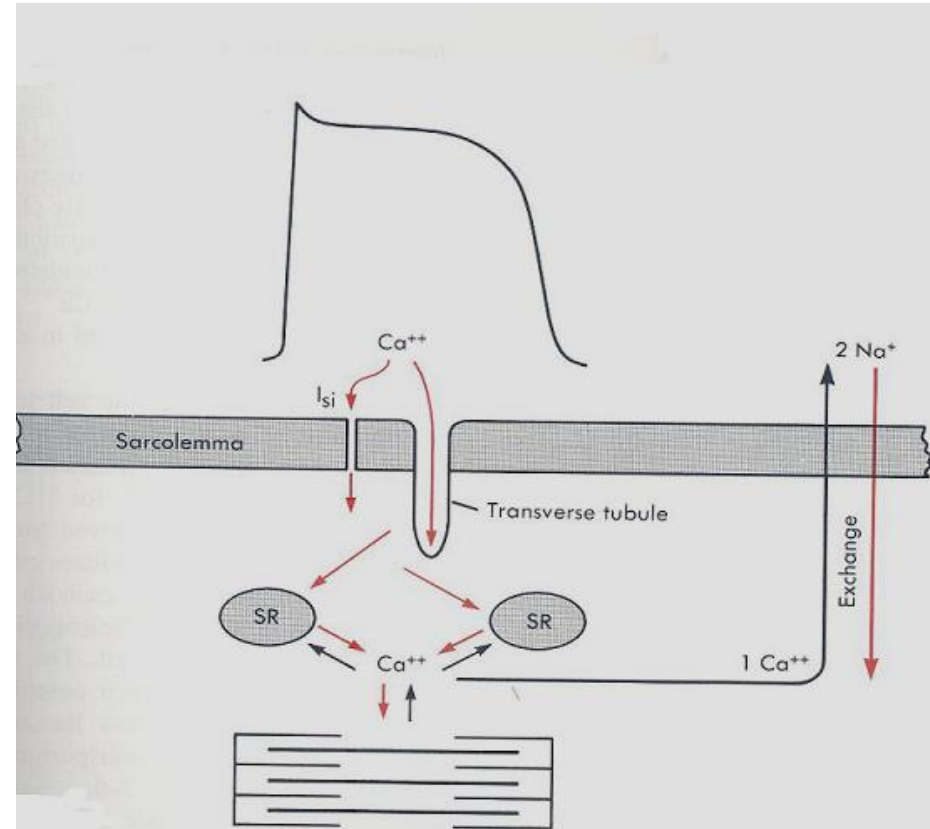
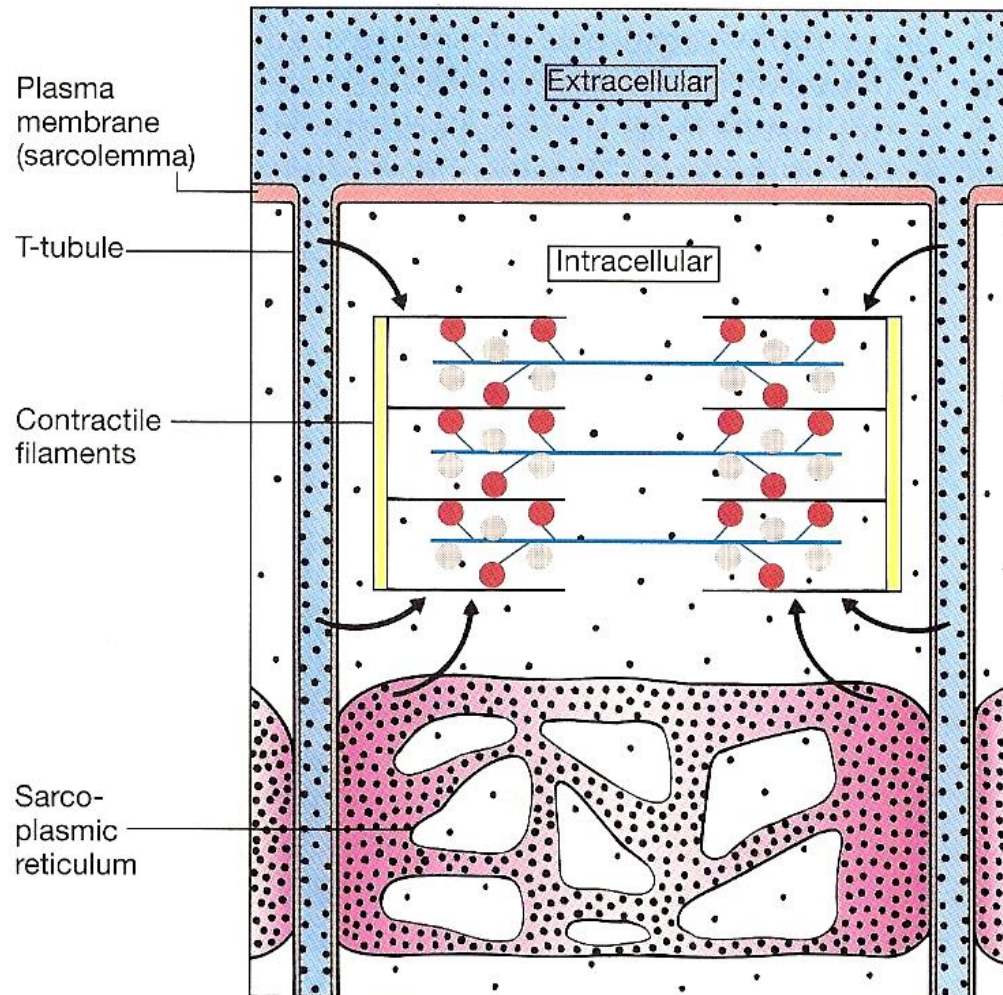
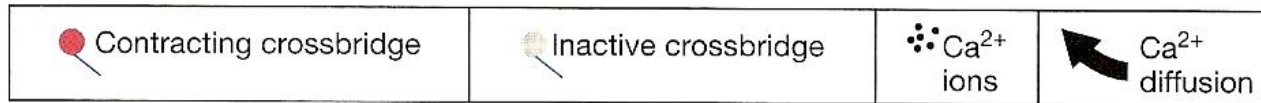


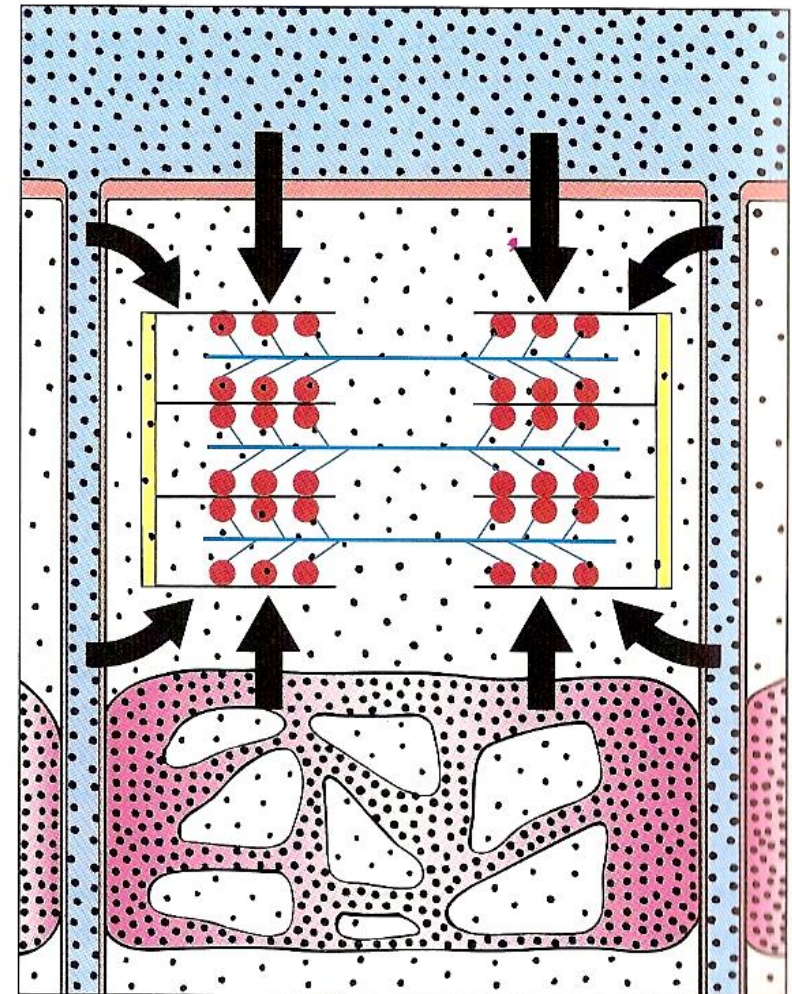
Fig. Excitation-contraction coupling in the heart

Excitation-Contraction Coupling



NORMAL CONTRACTILITY

(a)



INCREASED CONTRACTILITY

(b)

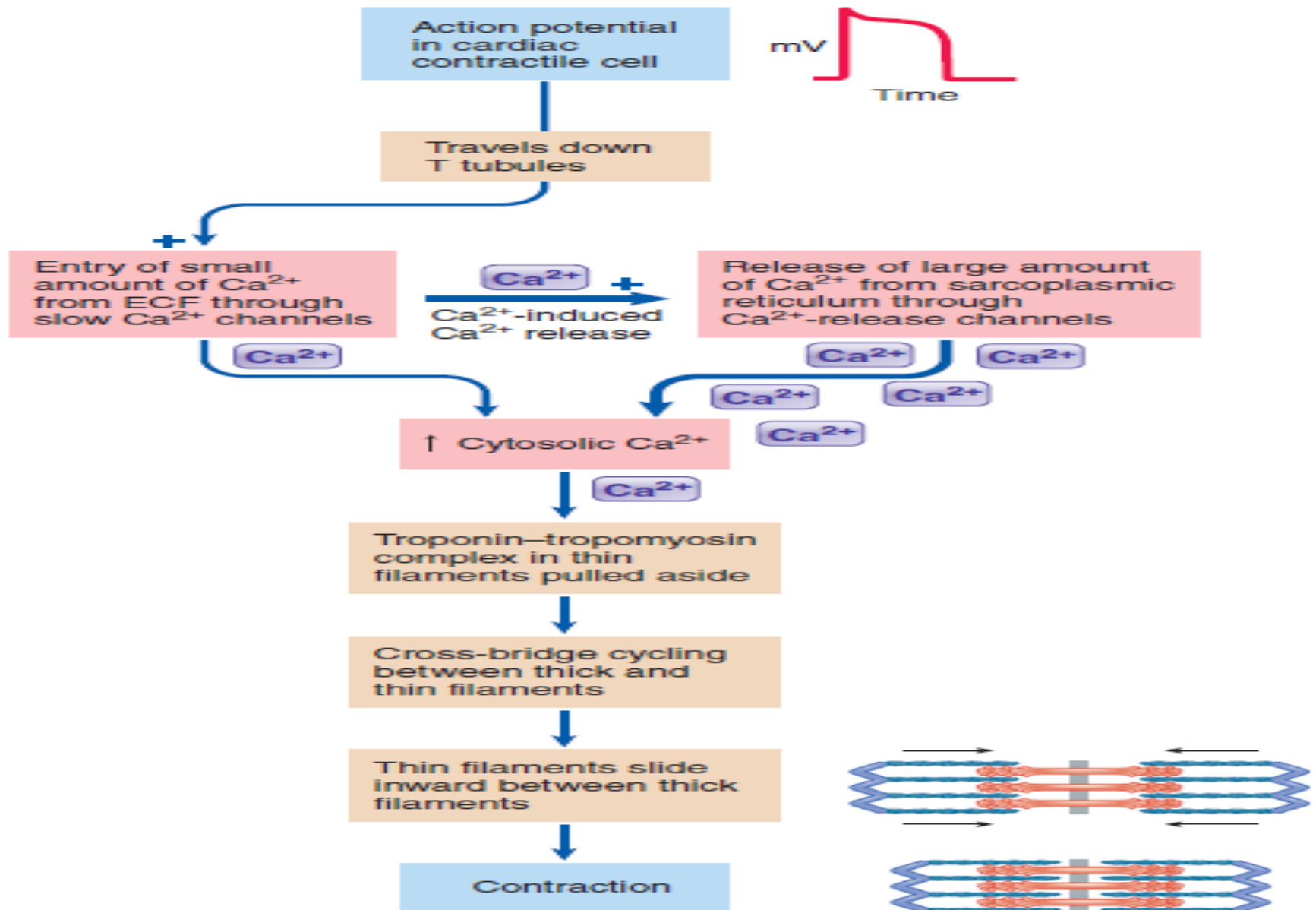
Contractility

- ♥ is the ability of the cardiac muscle to convert chemical energy into mechanical work.
- Myocardial fibers have '*Functional syncytium*' & NOT 'anatomical syncytium', because they present in contact but NOT in continuity.
- ♥ Strength of myocardial contraction determines the heart pumping power.
- ♥ Mechanism of contraction depends on the contractile filaments, which contain the protein molecules (actin & myosin).

Mechanism of Contraction

- The **mechanism of contraction** is similar in cardiac and skeletal muscle:
- The electrical activity leads to the mechanical response after a short delay.
- *As Ca^{2+} concentration rises inside a contractile fibre, Ca^{2+} binds to the regulatory protein troponin, which allows the actin and myosin filaments to begin sliding past one another, and tension starts to develop.*
- Substances that alter the movement of Ca^{2+} through slow Ca^{2+} channels influence the strength of heart contractions.
- **Epinephrine**, for example, increases contraction force by enhancing Ca^{2+} flow into the cytosol.
- **Calcium channel blocker** drugs decrease the heart contraction by decreasing Ca^{2+} flow into the cytosol.

Mechanism of Contraction



Factors affecting myocardial contractility

1. Cardiac Innervation:

- Sympathetic NS $\rightarrow \uparrow$ force of contraction.
- Parasympathetic NS (vagus) $\rightarrow \downarrow$ atrial force of contraction with no significant effect on ventricular ms.

2. Oxygen supply:

- Hypoxia $\rightarrow \downarrow$ contractility

3. Calcium & potassium ions concentration in ECF:

- $\uparrow \text{Ca}^{2+} \rightarrow \uparrow$ contractility
- $\uparrow \text{K}^+ \rightarrow \downarrow$ contractility

4. Physical factors:

- Warming $\rightarrow \uparrow$ contractility
- Cooling $\rightarrow \downarrow$ contractility

Factors affecting myocardial contractility

5. Hormonal & chemical factors (drugs):

- +ve inotropics:

(Adrenaline, noradrenaline, alkalosis, digitalis, Ca^{2+} , caffeine,...)

- -ve inotropics:

(Acetylcholine, acidosis, ether, chloroform, some bacterial toxins (e.g. diphtheria toxins), K^+ , ...)

Factors affecting myocardial contractility

6. Mechanical factors:

- a. Cardiac muscle obeys '*all or none law*':
i.e. minimal or threshold stimuli lead to maximal cardiac contraction, because cardiac muscles behaves as a syncytium.
- b. Cardiac muscle can't be stimulated while it is contracted, because its excitability during contraction is zero due to long ARP, so *it can't be tetanized*.
- c. Cardiac muscle can perform both *isometric & isotonic* types of contractions.

Factors affecting myocardial contractility

d. Starling's law of the heart:

■ “*Length-tension relationship*”

‘Within limits, the greater the initial length of the fiber, the stronger will be the force of its contraction;

➤ *However, overstretching the fiber as in heart failure its power of contractility decreases’*

i.e. within limits, the power of contraction is directly proportional to the initial length of the muscle.

- Cardiac muscle accommodates itself (up to certain limit) to the changes in venous return.

Factors affecting myocardial contractility

- e. Cardiac muscles shows *staircase phenomenon (gradation)*, if providing all other conditions kept constant.

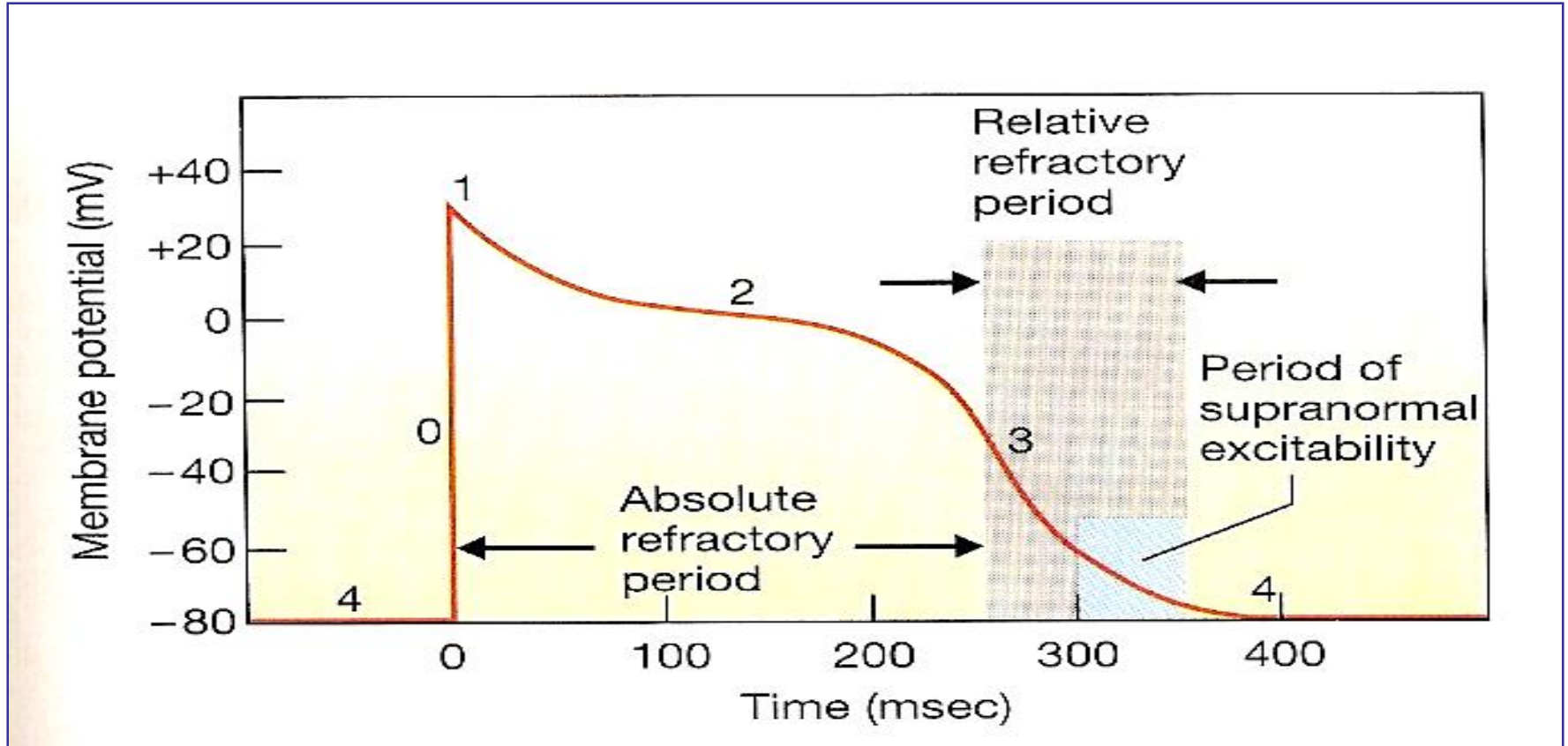
i.e. if an isolated heart is stimulated by successive equal & effective stimuli, the 1st few contractions show a gradual
↑ in the magnitude of contraction.

Refractoriness in the Contractile Fibers

- ❖ During the phase of contraction, the heart is unexcitable (refractory) to stimulation.
- ❖ There are two degrees of refractoriness:
 - *Relative refractoriness*: Only APs of strong stimuli can be generated.
 - *Absolute refractoriness*: No other AP can be triggered, even by extremely strong stimuli.
 - Because of this refractoriness, the heart muscle cannot be tetanized.

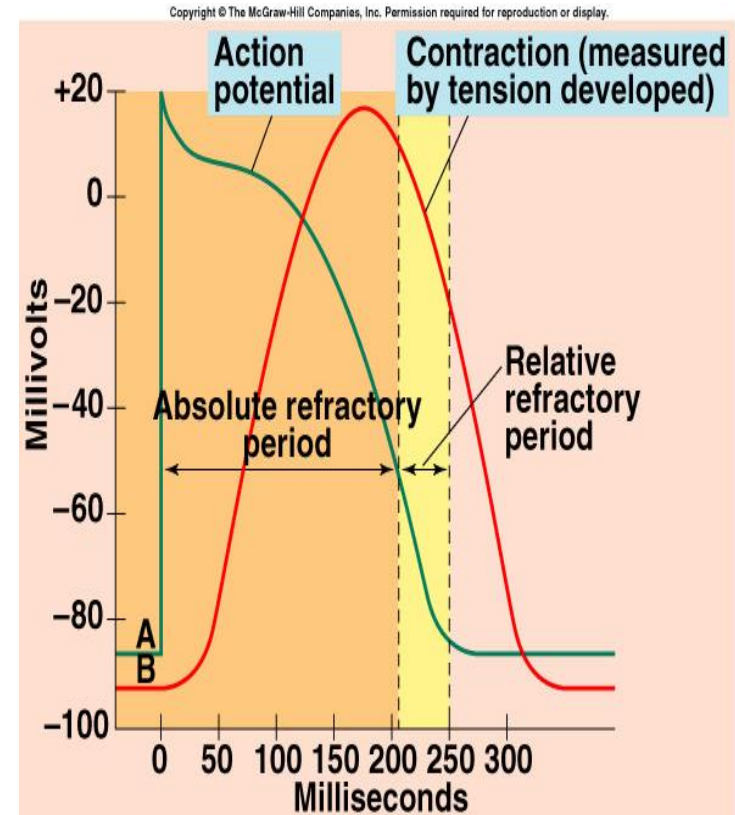
Refractoriness

- Refractoriness passes through three periods:
 1. Absolute refractory period (ARP)
 2. Relative refractory period (RRP)
 3. Dangerous period (supranormal period)



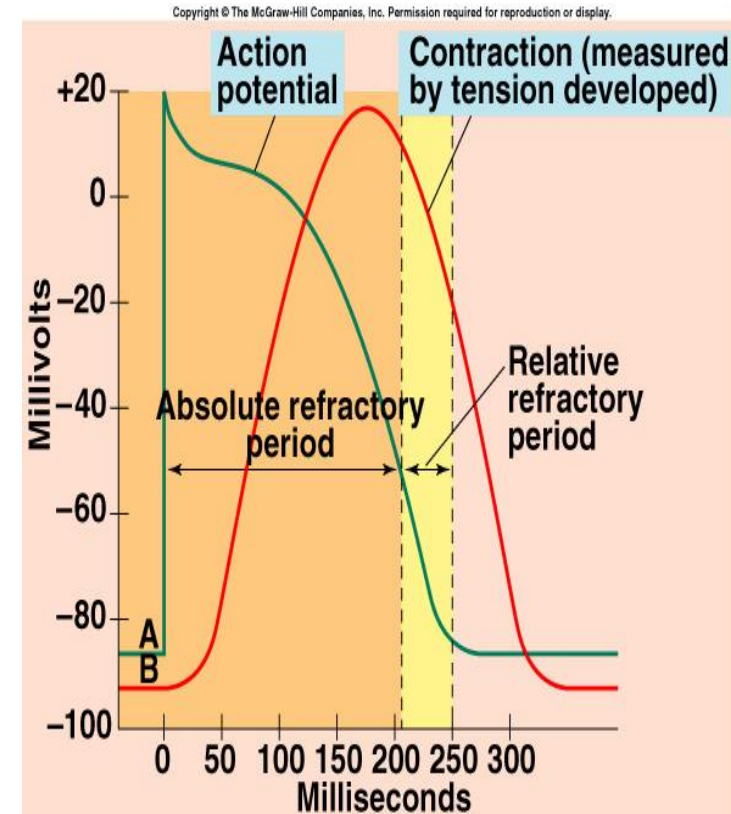
1. Absolute Refractory Period (ARP):

- The excitability of cardiac muscle is completely lost during this period, i.e. doesn't respond to 2nd stimulus.
- Very long.
- Occupies the whole period of systole.
- Corresponds to the period of depolarization (phase 0), & the first two phases of repolarization.
- Heart can't be tetanized, as its ARP occupies the whole contraction phase.



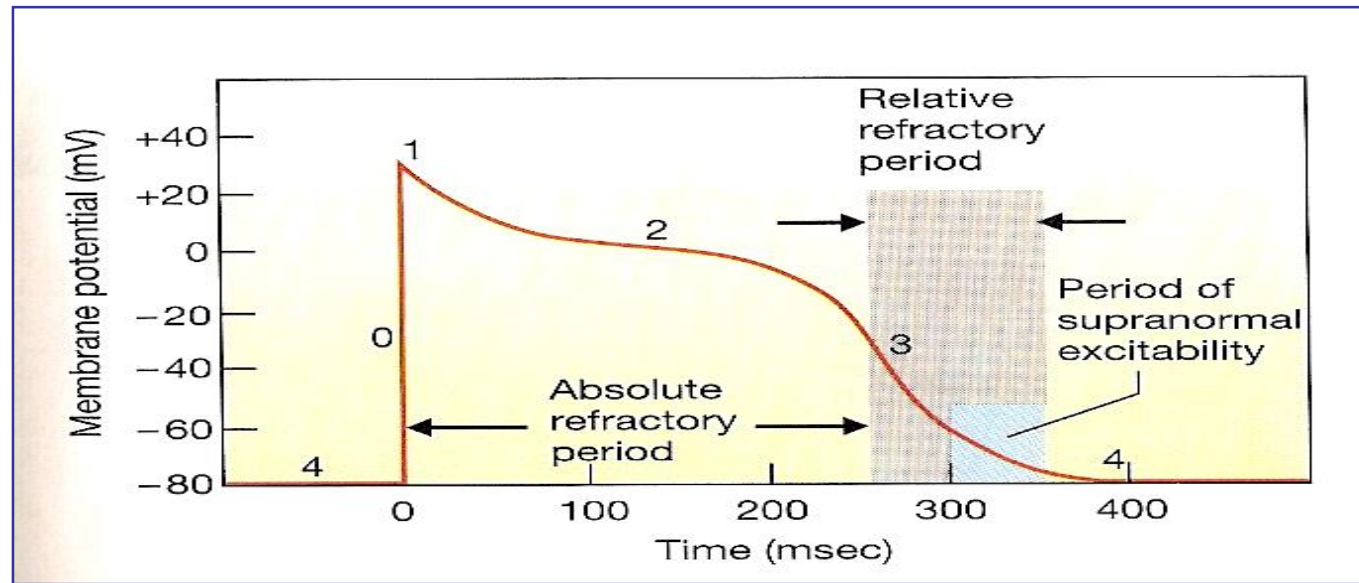
2. Relative Refractory Period (RRP):

- The excitability of cardiac muscle is partially recovered during this period, i.e. stronger stimuli than normal are required to excite the muscle.
- Occupies the time of diastole.
- Corresponds to the 3rd phase of repolarization.
- Can be affected by the HR, temp., bacterial toxins, vagal stimulation, sympathetic stimulation & drugs.



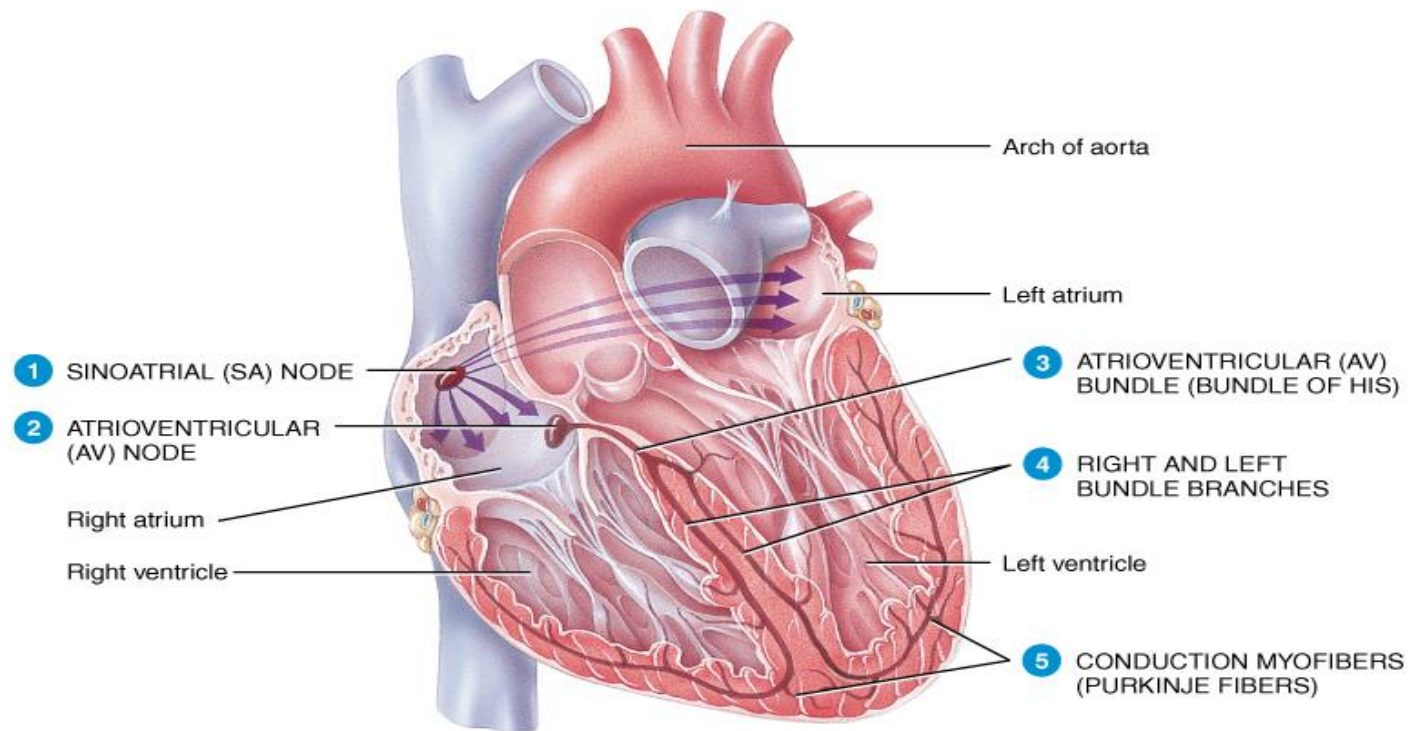
3. Dangerous Period (Supranormal):

- The excitability of cardiac muscle is supranormal just at the end of the AP, i.e. *weaker stimuli than normal can excite the muscle.*



Conduction of Heart

- Each action potential from the SA node propagates throughout both atria **via gap junctions** in the intercalated discs of atrial muscle fibres.

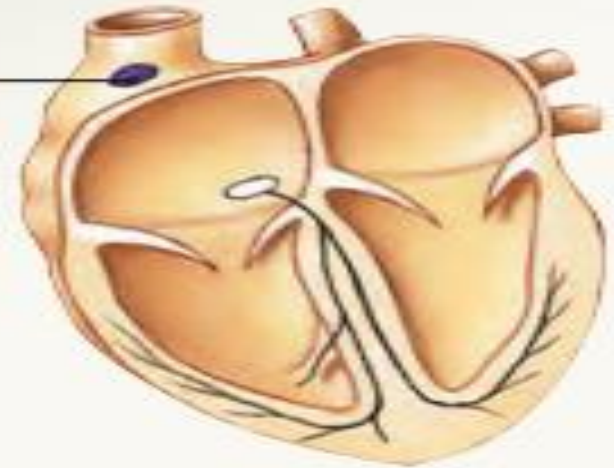


Conduction of heart

1

SA node activity and atrial activation begin.

SA node

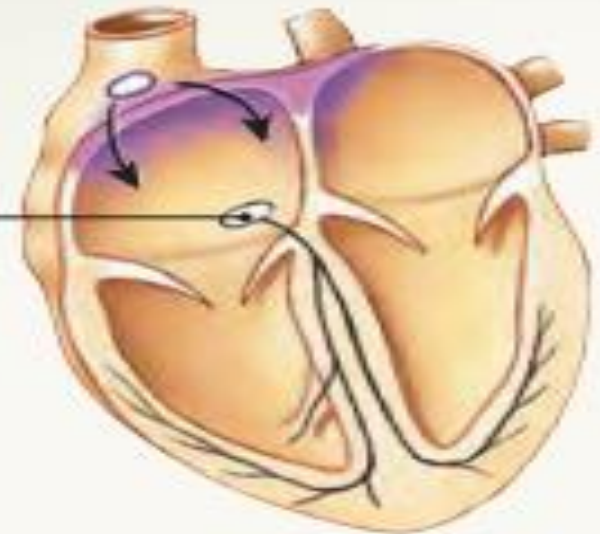


Time = 0

2

Stimulus spreads across the atrial surfaces and reaches the AV node.

AV node



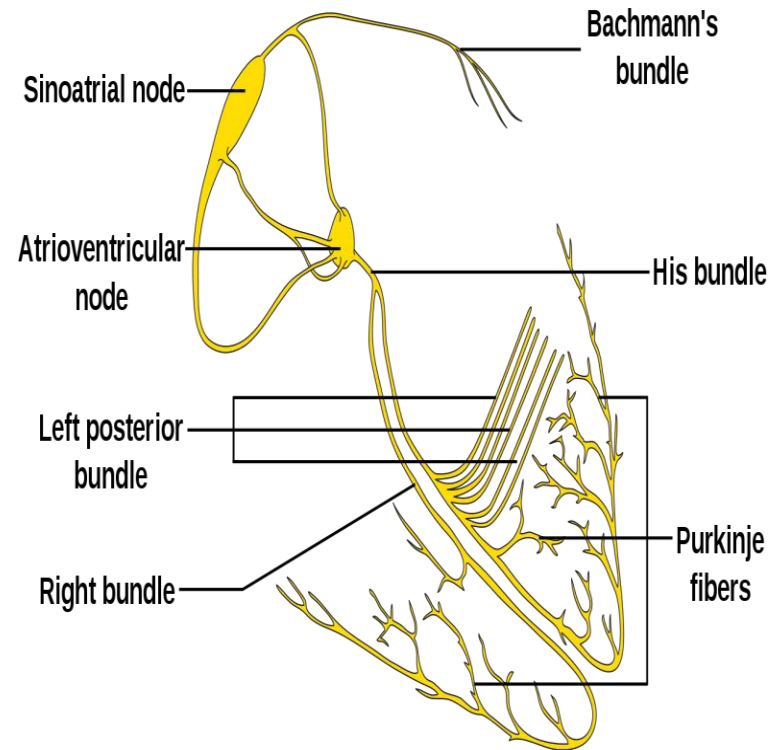
Elapsed time = 50 msec

Conduction of heart

- By conducting along atrial muscle fibres, the action potential reaches the **atrioventricular (AV) node**, located in the **interatrial septum**, just anterior to the opening of the coronary sinus from the left side of heart.
- The SA node is connected to the larger AV node **by three bundles of atrial fibres** that contain Purkinje type fibres.
- Such as
 - *The anterior internodal tract,*
 - *The middle internodal tract,*
 - *The posterior intermodal tract.*
- Conduction also occurs through atrial myocytes, but it is more rapid in these bundles.

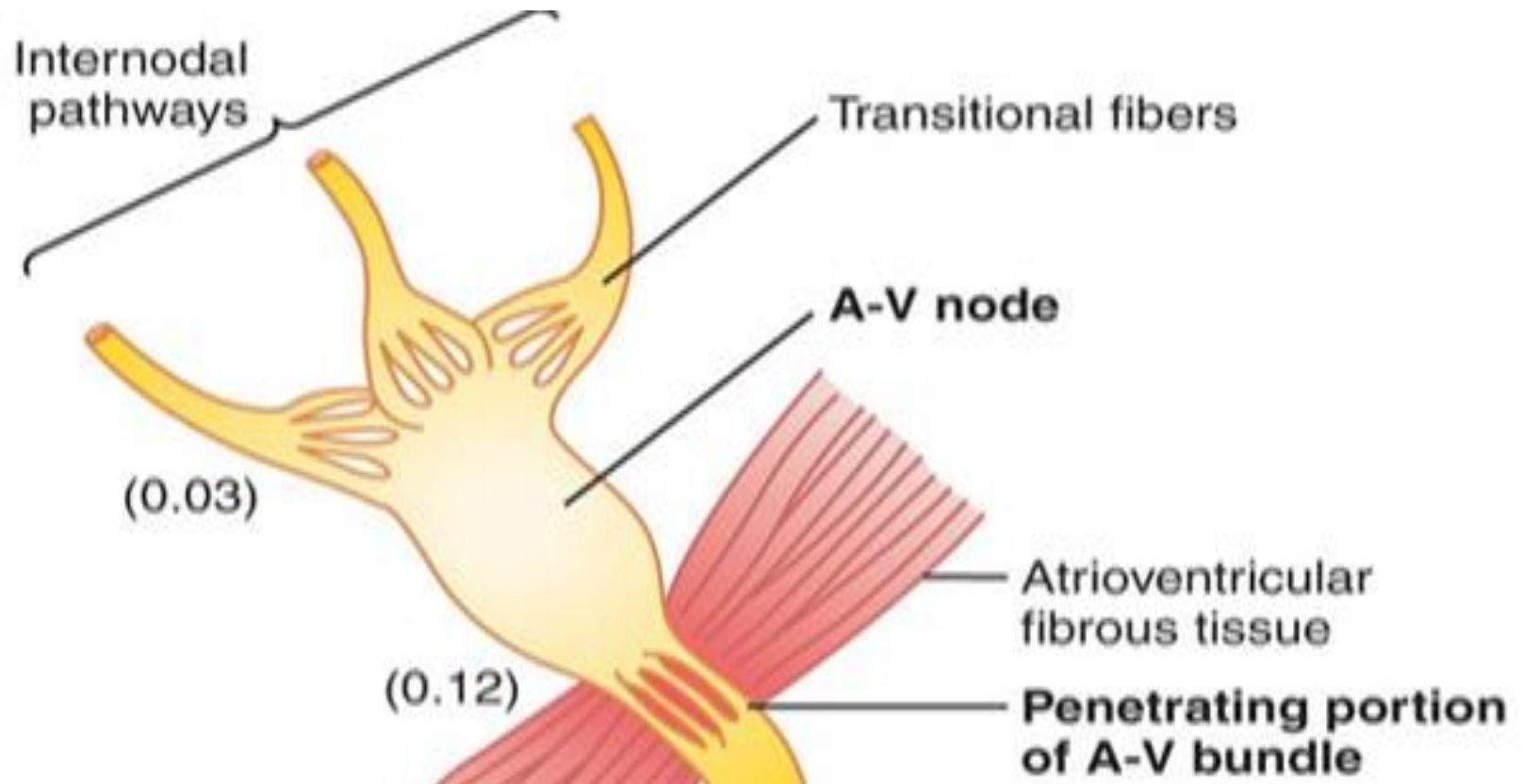
Bachmann's Bundle

- is a branch of the anterior internodal tract that passes from the right atrium and resides on the inner wall of the left atrium.
- During normal sinus rhythm, it is the preferential path for electrical activation of the left atrium.
- It is therefore considered to be part of the "atrial conduction system" of the heart.



Conduction of heart

- An action potential takes approximately *30-50 msec* to travel from the SA node to the AV node along these pathways.



Conduction of heart

- At the AV node, the action potential **slows** considerably as a result of various differences in cell structure in the **AV node**.
- Such as:
 - AV-node contains *many myocardial cells with small diameter*.
 - Contain *very less no of gap junction* between the cells.
 - RMP is -50mv which is less in it's negativity but it is *poor in the attraction of depolarizing ions*.
 - Is dependent on *slowly acting calcium ion channels* for depolarization.

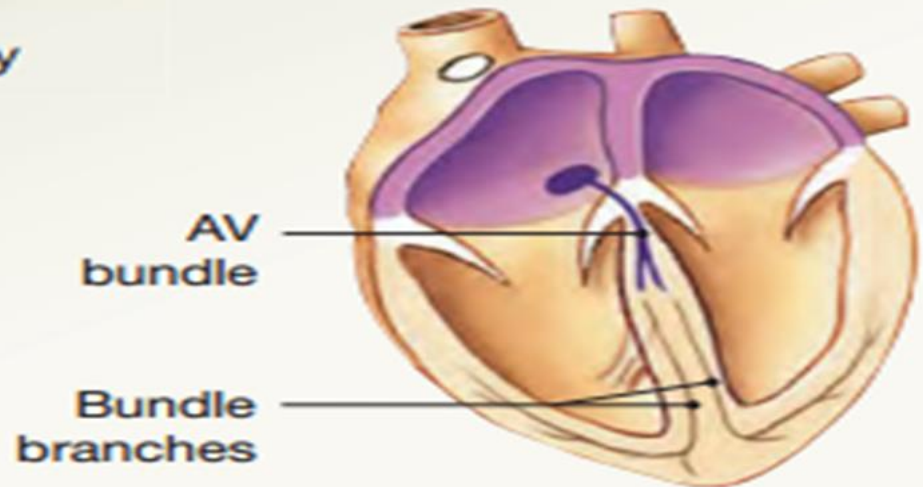
Conduction of heart

- *This delay provides time for the atria to contract and empty their blood into the ventricles before the ventricles undergo depolarization.*
- An important property that is unique to the AV node is **decremental conduction**, in which *the more frequently the node is stimulated the slower it conducts.*
- This is the property of the AV node that prevents rapid conduction to the ventricle in cases of rapid atrial rhythms, such as atrial fibrillation or atrial flutter.

Conduction of heart

3

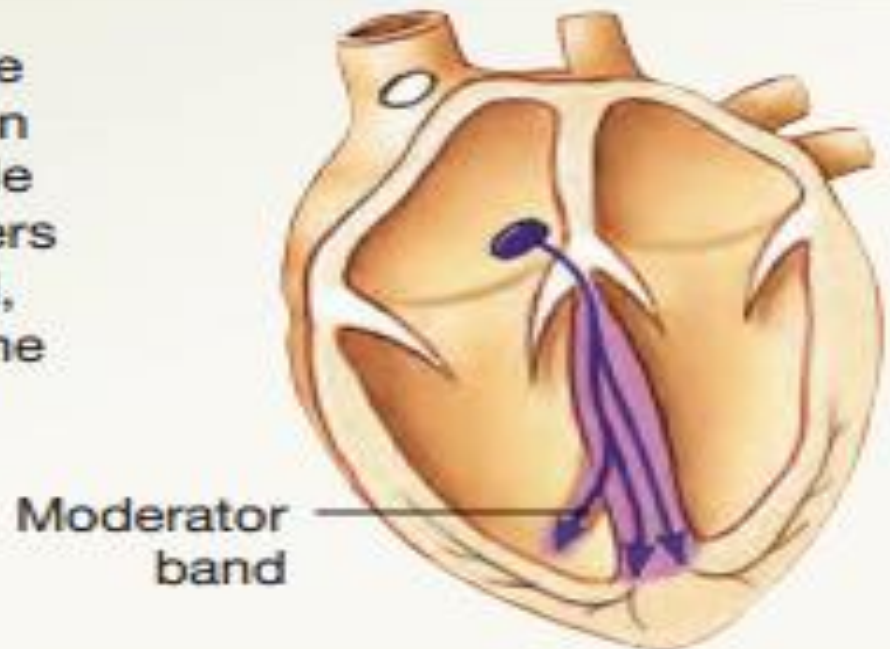
There is a 100-msec delay at the AV node. Atrial contraction begins.



Elapsed time = 150 msec

4

The impulse travels along the interventricular septum within the AV bundle and the bundle branches to the Purkinje fibers and, via the moderator band, to the papillary muscles of the right ventricle.



Elapsed time = 175 msec

Conduction of heart

- From the AV node, the action potential enters the atrioventricular (AV) bundle OR (bundle of His).
- *This bundle is the only site where action potentials can conduct from the atria to the ventricles.*
- After propagating along the AV bundle, the action potential enters both the right and left bundle branches.
 - ***The left bundle branch***, which supplies the massive left ventricle, is much larger than the right bundle branch.
- The bundle branches extend through the interventricular septum toward the apex of the heart and fan out deep to the endocardial surface by **Purkinje fibres**.

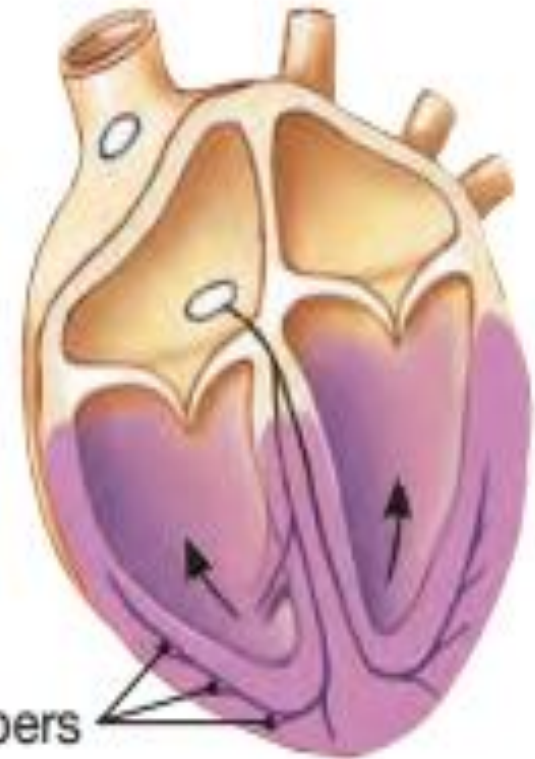
Conduction of heart

5

The impulse is distributed by Purkinje fibers and relayed throughout the ventricular myocardium. Atrial contraction is completed, and ventricular contraction begins.

Elapsed time = 225 msec

Purkinje fibers



Velocity of conduction of AP in cardiac muscles

- The slow conduction in the internodal pathways, AV nodal, and AV bundle fibers is caused mainly by diminished numbers of gap junctions between successive cells in the conducting pathways

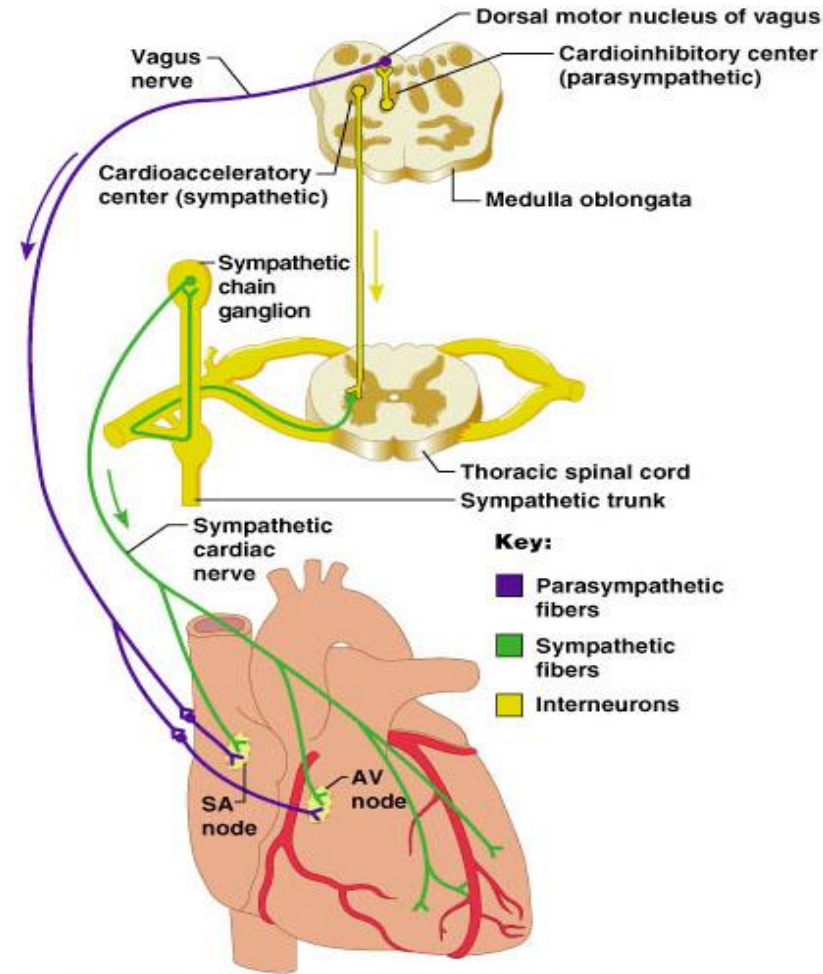
Structures	Conduction velocity (m/s)
SA-node	0.05
Internodal fibers	1.0
Atrial muscle	0.3
AV-node	0.05
AV-bundle	1.0
Purkinje fibres	4.0
Ventricular muscles	1.0

Factors Affecting Autorhythmicity

- **Sympathetic nerve** ↑ autorhythmicity, + ve chronotropic effect
- **Parasympathetic nerve** :Inhibits autorhythmicity, -ve chronotropic effect.
- **Temperature**: ↑BT by 1°C, ↑HR by 20 beats/min
 - ↑BT increases SA- nodal discharge
- **Drugs**: adrenalin, noradrenalin, T3/T4 increase cardiac rhythmicity they have +ve chronotropic effect
- **Cholinergic drugs**: Ach, pilocarpin→ -ve chronotropic effect
- **Electrolytes**: Hyperkalemia ($[K^+]$ 5-9 meq/L) leads to bradycardia by hyperpolarizing the membrane.
- **Mechanical factors**: Atrial wall stretch by ↑blood volume, ↑HR

Heart Innervation

- The heart is supplied with both sympathetic and parasympathetic nerves.
 - Dual autonomic innervation
- The sympathetic nerve and parasympathetic nerve supply to the heart is controlled by:
 - The medullary vasoconstrictor/ cardio accelerator center and
 - The medullary vasodilator/ cardio inhibitor center called the **nucleus of the tractus solitaries**.



Heart Innervation

Sympathetic and parasympathetic receptors and their functions in the heart and vessels

		Heart			Vessels	
		Receptor	Function			Receptor Function
			Inotropy	Chronotropy	Dromotropy	
Norepinephrine	α_1	+		+	+	α_1 Vasoconstriction
	β_1	+		+	+	β_1 Vasoconstriction
	β_2	+		+	+	β_2 Vasodilation
Acetylcholine	M ₂	-		-	-	M ₂ Vasodilation

Heart Innervation- Parasympathetic Nerves

- The vagus nerve has 3 nuclei in the CNS associated with cardiovascular control:
 - The nucleus ambiguus, the dorsal motor nucleus, and the solitary nucleus.
- The parasympathetic output to the heart comes mainly from neurons in the *nucleus ambiguus* and to a lesser extent from the dorsal motor nucleus
- The parasympathetic nerves (**the vagus**) are distributed mainly to the SA and AV nodes, to a lesser extent to the muscle of the two atria, and *very little (almost none) to the ventricular muscle*.
 - The right vagus has a strong influence on SA node, while the left vagus has dominant effect on AV node.

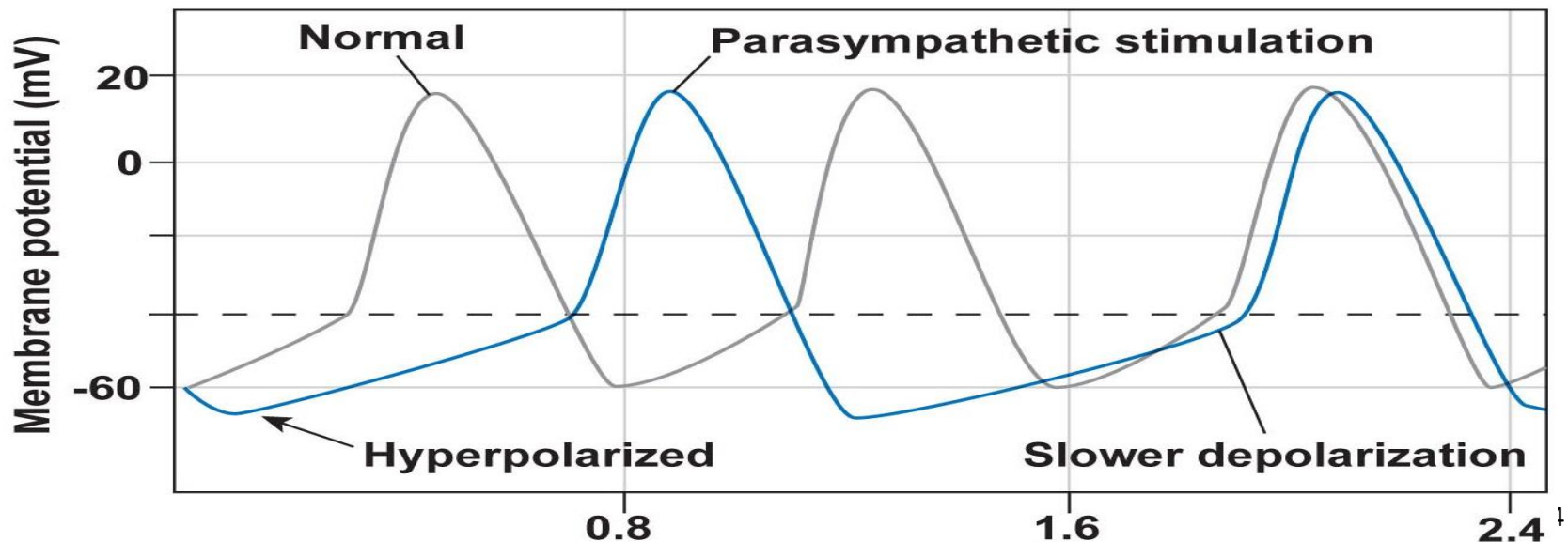
Heart Innervation- Parasympathetic Nerves

- Stimulation of the parasympathetic nerves to the heart (the vagus) causes the acetylcholine to be released at the vagal endings and activates *muscarinic receptors*.
 - First, it decreases the rate of rhythm of the sinus node.
 - *By increasing the permeability of the fiber membranes to potassium ions*, which allows rapid leakage of potassium out of the conductive fibres (*hyperpolarization*).
 - Second, it *decreases the excitability of the AV junctional fibres* between the atrial musculature and the AV node, thereby slowing transmission of the cardiac impulse into the ventricles.

Heart Innervation- Parasympathetic Nerves

Parasympathetic Activity Summary:

- Decreased chronotropic effects
↓ heart rate
- Decreased dromotropic effects
↓ conduction of APs
- Decreased inotropic effects
↓ contractility



Heart Innervation- Parasympathetic Nerves

- Weak to moderate **vagal stimulation** slows the rate and strong stimulation of the vagi can stop completely the rhythmical excitation of the sinus node into the ventricles.
- But some small area in the **Purkinje fibres**, usually in the ventricular septal portion of the AV bundle, develops a rhythm of its own and causes ventricular contraction at a rate of 15 to 40 beats per minute.
 - This phenomenon is called **ventricular escape**.

Heart Innervation- Sympathetic Nerves

- The sympathetic nerves, **conversely distributed to all parts of the heart**, with strong representation to the ventricular muscle, as well as to all the other areas.
- Sympathetic stimulation causes essentially the opposite effects on the heart to those caused by vagal stimulation, as follows:
 - First, it increases the *rate of sinus nodal discharge*.
 - Second, it increases the *rate of conduction*, as well as the level of excitability in all portions of the heart.
 - Third, it increases greatly the *force of contraction* of all the cardiac musculature, both atrial and ventricular

Heart Innervation- Sympathetic Nerves

- Stimulation of the sympathetic nerves releases the hormone *norepinephrine* at the sympathetic nerve endings.
- Norepinephrine in turn stimulates *beta-1 adrenergic receptors*, which mediate the effects on heart rate.
- Activation of beta-1 adrenergic receptors increases the permeability of the fibre membrane to *sodium and calcium ions*.
- An increase of sodium-calcium permeability causes a more positive resting potential and also causes increased rate of self-excitation.
 - Therefore, increasing the heart rate.

Heart Innervation- Sympathetic Nerves

Sympathetic Activity Summary:

Increased chronotropic effects

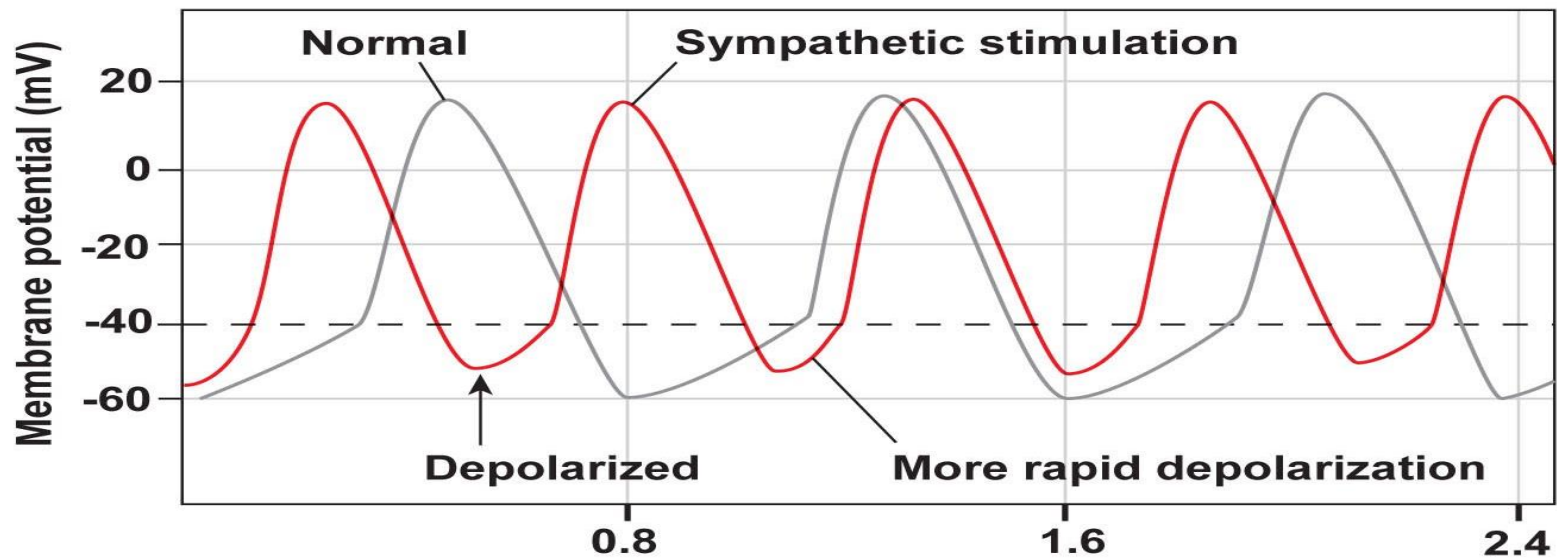
↑ heart rate

Increased dromotropic effects

↑ conduction of APs

Increased inotropic effects

↑ contractility



Abnormal Pacemakers-"Ectopic" Pacemaker

- Occasionally some other part of the heart develops a rhythmical discharge rate that is *more rapid than that of the sinus node*.
- For instance, sometimes occurs in the AV node or in the Purkinje fibres when one of these becomes abnormal.
 - In either case, the pacemaker of the heart shifts from the sinus node to the A-V node or to the excited Purkinje fibers.
- Under rarer conditions, a place in the atrial or ventricular muscle develops excessive excitability and becomes the pacemaker.
 - *A pacemaker elsewhere than the sinus node is called an "ectopic" pacemaker.*
 - An ectopic pacemaker causes an abnormal sequence of contraction of the different parts of the heart and can cause significant weakness of heart pumping.